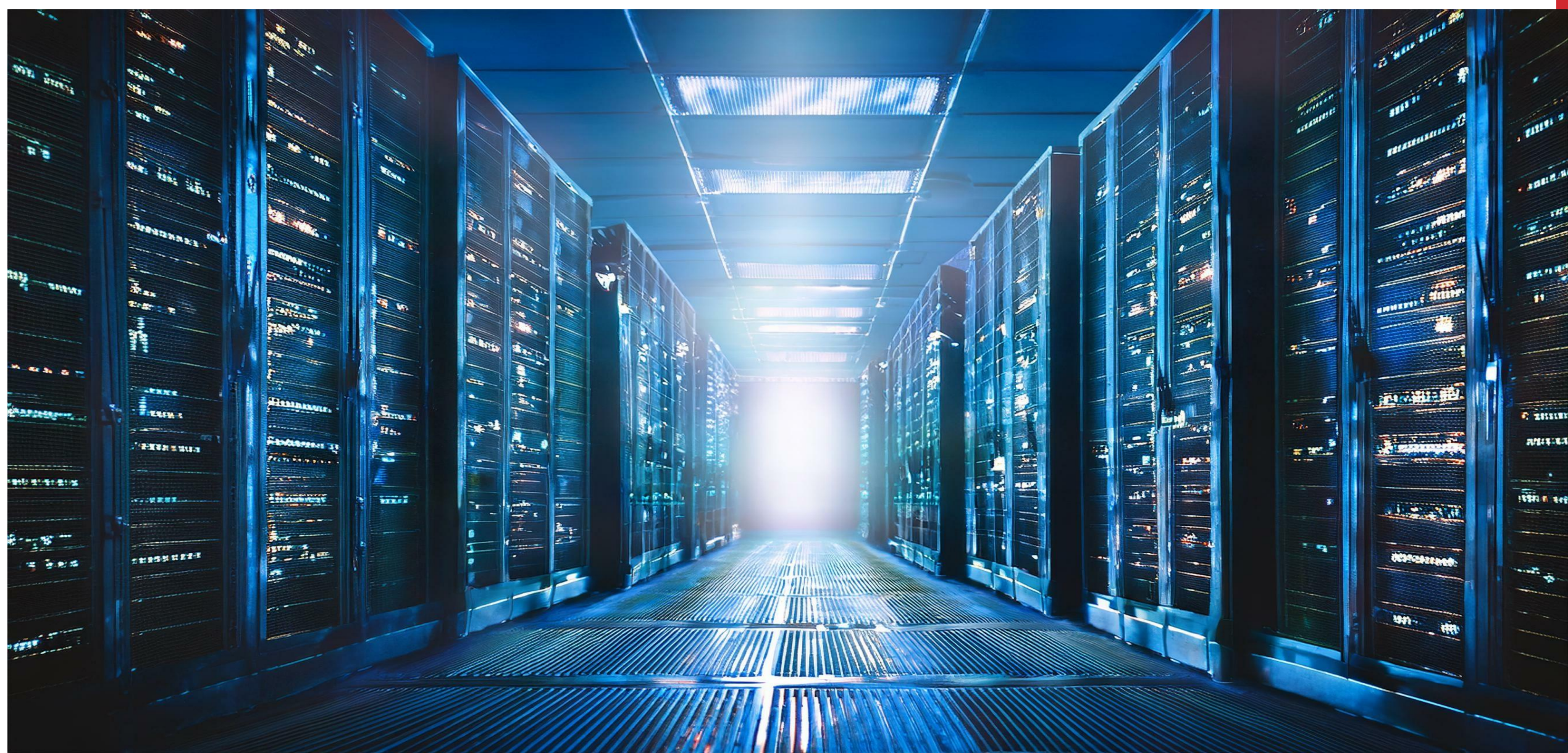


Data Protection & Management



LOGISTICS

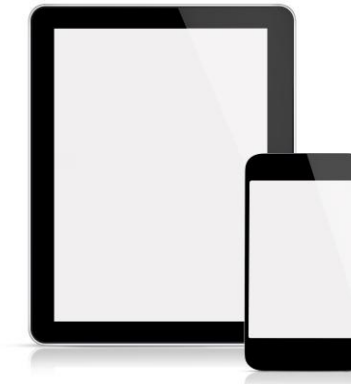
Class Hours:

- Class starts at X:00 AM/PM Time and Finish at XX:00 AM/PM
- There will be regular breaks in class.



Telecommunication:

- Turn off or set electronic devices to silent (not vibrate)
- Reading or attending to devices can be distracting to other students
- Try to delay until breaks or after class



COURSE STRUCTURE

Day 1: Data Fundamentals

Day 2: Flash Storage & Pure Storage

Day 3: Data Protection & Management

Day 4: Kubernetes Integration & Modern Applications

PREREQUISITES & EXPECTATIONS

- Basic understanding of computer systems, networking, and programming fundamentals
- One or more years of technical experience

AGENDA FOR DAY 3

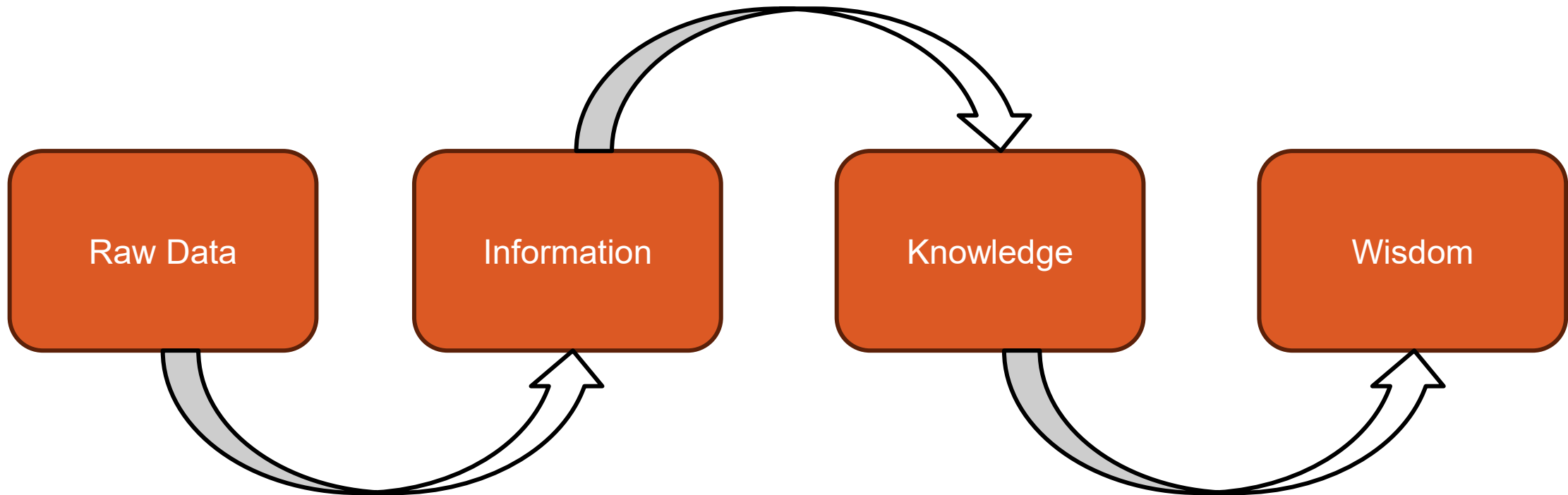
1. Data Fundamentals
2. Data Protection
3. Disaster Recovery & Failover



DATA FUNDAMENTALS

WHAT IS DATA

- Definition : In the IT context, the term data represent facts, that can be collected, stored and processed.



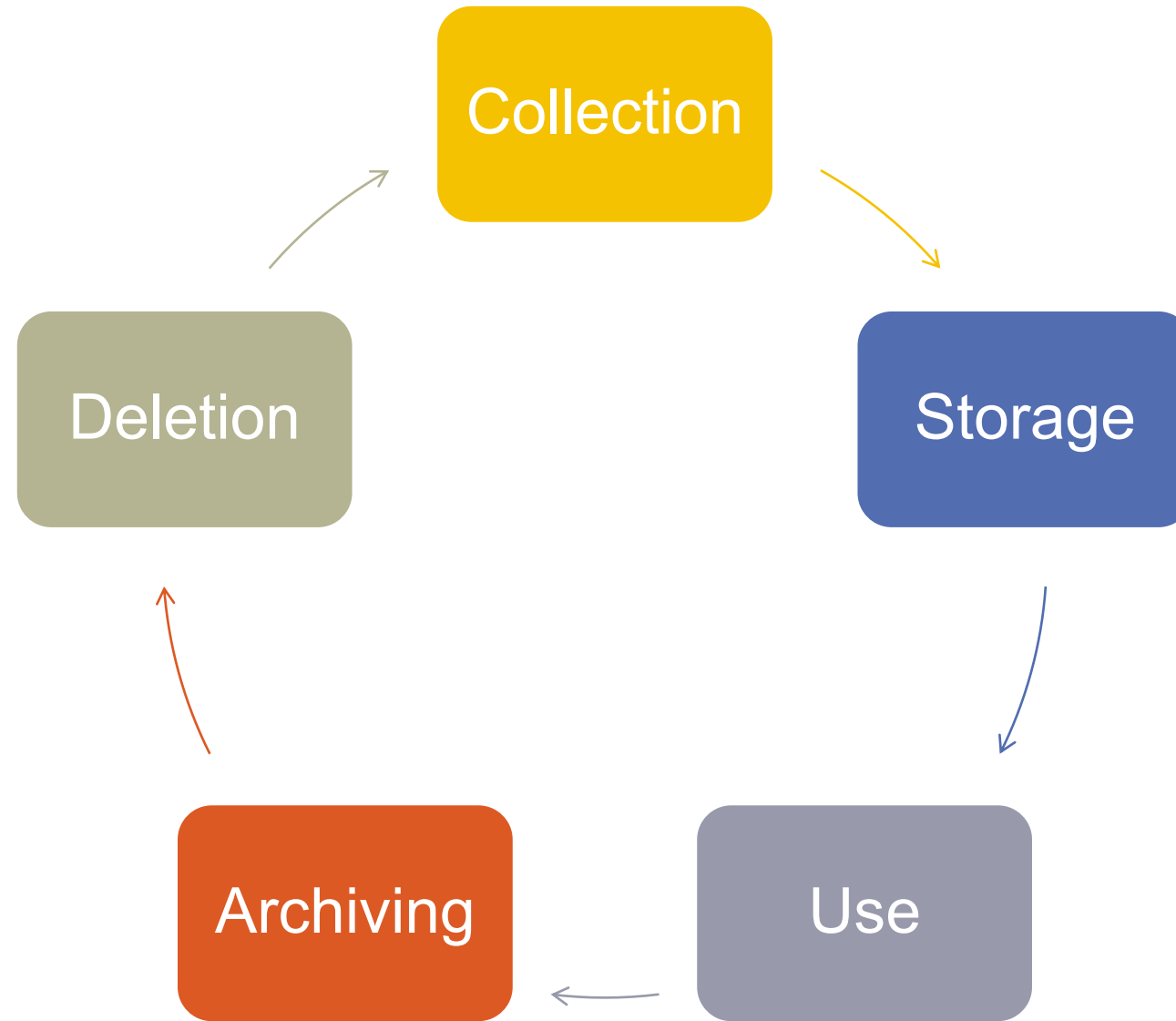
DATA MANAGEMENT

- Data is a valuable asset
- Data Management is cross-functional
- Data Management requires an holistic approach
- Data Management is in charge of the data lifecycle
- Effective Data Management requires leadership commitment

WHY DATA MATTERS

- Enables data-driven decision-making
- Powers automation and innovation
- Supports customer personalization
- Must be protected to maintain trust

THE DATA LIFECYCLE



PRINCIPLES OF RESPONSIBLE DATA HANDLING

- Collect only necessary data
- Store it securely
- Use it lawfully and fairly
- Retain it for limited time
- Dispose safely and permanently

WHO OWNS THE DATA?

- Collect only necessary data
- Store it securely
- Use it lawfully and fairly
- Retain it for limited time
- Dispose safely and permanently

DATA QUALITY DIMENSIONS

- Accuracy: Data reflects reality
- Consistency: Uniform across systems
- Completeness: Nothing essential is missing
- Timeliness: Updated when needed

INTRODUCTION TO GDPR

GDPR = *General Data Protection Regulation (EU, 2018)*

Applies to anyone handling EU personal data.

Focuses on **privacy, fairness, and accountability**.

Scope :

EU citizens' personal data

Controllers and processors

Digital and paper records alike

KEY GDPR ROLES

Term	Definition	Example
Data Subject	Person whose data is processed	Customer
Controller	Decides how and why data is used	Company
Processor	Handles data for controller	Cloud vendor
DPO	Ensures compliance	Data Protection Officer

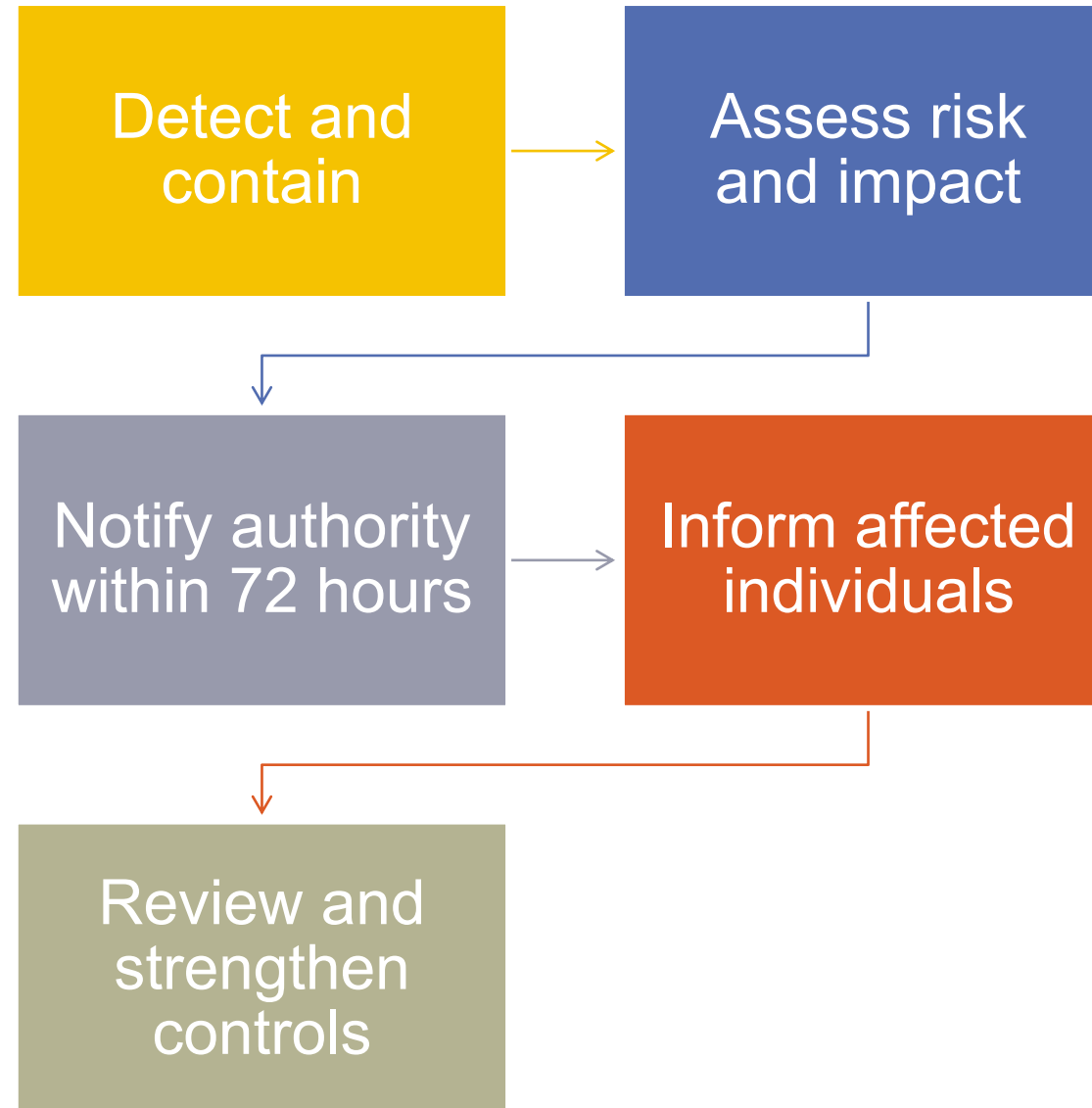
THE 7 GDPR PRINCIPLES

Principle	Meaning
1. Lawfulness, Fairness, Transparency	Clear and honest processing
2. Purpose Limitation	Use data only for the reason collected
3. Data Minimization	Collect only what is necessary
4. Accuracy	Keep data correct and updated
5. Storage Limitation	Delete when no longer needed
6. Integrity & Confidentiality	Protect with security controls
7. Accountability	Demonstrate compliance

DATA CLASSIFICATION

Level	Description	Example
Public	Accessible to everyone	Website content
Internal	Limited to employees	Process manuals
Confidential	Restricted	HR records
Sensitive	Special protection required	Financial or health data

MANAGING DATA BREACHES



BUILDING A DATA-RESPONSIBLE CULTURE

Awareness



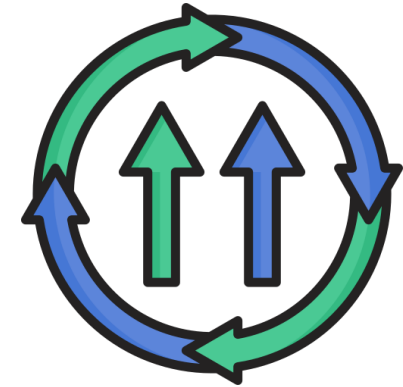
Accountability



Ethics



Continuous Improvements



POP QUIZ:

What are the components of a network ?

- A. Endpoints
- B. Interstations
- C. Intermediaries
- D. Hosts



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DATA PROTECTION

THE COST OF DATA LOSS

- Sobering Statistics:
 - Average data breach cost: \$4.45M (2023)
 - 60% of small businesses close within 6 months of data loss
 - Ransomware attacks every 11 seconds globally
- Business Impact Examples:
 - TSB Bank (2018): £330M+ losses from IT failure
 - British Airways (2017): £183M fine for data breach
 - Maersk (2017): \$300M+ losses from NotPetya attack
- Recovery Time Statistics:
 - 25% of companies take >1 week to restore operations
 - Each hour of downtime costs \$300K+ for enterprise

DATA PROTECTION BEST PRACTICES

- 3-2-1 Backup Rule:
 - 3 copies of critical data / 2 different storage media types / 1 offsite copy
- Modern Enhancements:
 - 3-2-1-1: Add immutable/air-gapped copy
 - 3-2-1-0: Zero tolerance for unverified backups
- Testing and Validation:
 - Regular restore testing
 - Automated backup verification
 - Disaster recovery exercises
- Documentation and Training:
 - Updated recovery procedures
 - Staff training and certification
 - Vendor contact information and escalation

LAYER OF DATA PROTECTION

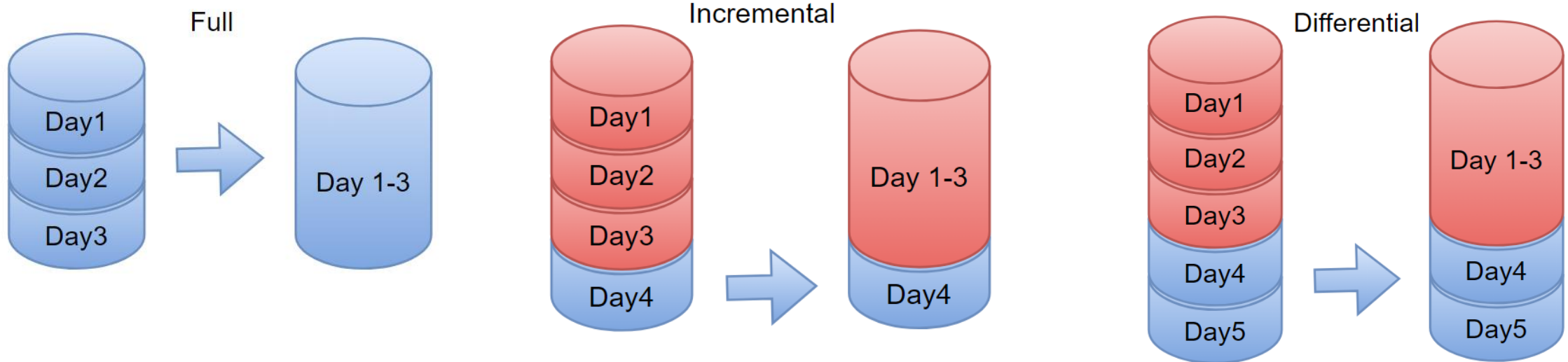
	Layer	Goals	Examples
	Preventive	Avoid Data Loss	Access controls, encryption
	Detective	Identify Incidents	Monitoring, alerting
	Corrective	Recover from damage	Backup, Snapshot, replication

THE COST OF DATA LOSS

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BACKUP – THE FOUNDATION OF RECOVERY

- Copy of data stored separately from production
- Used for recovery after loss or corruption
- Stored on-premises, offsite, or in the cloud
- Can be full, incremental, or differential



UNDERSTANDING BACKUP STRATEGIES

Backups aren't one-size-fits-all.

Each backup strategy must balance **cost, performance, and recovery speed**.

Three main approaches:

Hot backup

Cold backup

Warm backup



HOT BACKUP

Backup performed while the system is running.

No downtime – users continue accessing data.

Requires advanced tools (e.g., database shadow copies, transaction logs).

Ideal for mission-critical systems.

COLD BACKUP

System is offline during backup.
Users cannot access data during the process.
Simple, reliable, but slowest to recover.

Best for:

Non-critical data

Archival storage

Systems with predictable downtime

COMPARISON – HOT VS WARM VS COLD

Feature	Hot Backup	Warm Backup
System State	Online & active	Standby, ready to start
Data Synchronization	Real-time	Periodic
Downtime	None	Short
Cost	High	Medium
Typical Use	Banking, healthcare, e-commerce	CRM, HR systems

BACKUP STRATEGY AND BUSINESS CONTINUITY

- Match backup type to Recovery Time Objective (RTO) and Recovery Point Objective (RPO)
- Regularly test restoration procedures
- Combine hot, warm, and cold tiers for cost efficiency
- Document roles, timing, and tools

SNAPSHOT – POINT-IN-TIME PROTECTION

- A snapshot is a frozen image of data at a specific moment. It captures state instantly — with minimal performance impact.
- Use Cases:
 - Fast rollback (accidental deletion, corruption)
 - Pre-update checkpoints
 - Cloud volume protection
 - Production snapshot

SYNCHRONOUS REPLICATION

- Characteristics:
 - Writes acknowledged after remote confirmation
 - Zero data loss (RPO = 0)
 - Distance limitations (< 100km typically)
- Use Cases:
 - Mission-critical applications
 - Regulatory compliance requirements
 - Active-active clustering
- Performance Impact:
 - Latency increases write response time
 - Network bandwidth requirements
 - Quality of service considerations

ASYNCHRONOUS REPLICATION

- Characteristics:
 - Writes acknowledged locally, replicated later
 - Minimal data loss (RPO = minutes)
 - Unlimited distance capability
- Implementation Methods:
 - Periodic: Scheduled replication intervals
 - Continuous: Real-time change shipping
 - Change Log: Transaction log-based replication
- Advantages:
 - Lower performance impact
 - Higher bandwidth efficiency
 - Cost-effective for long distances

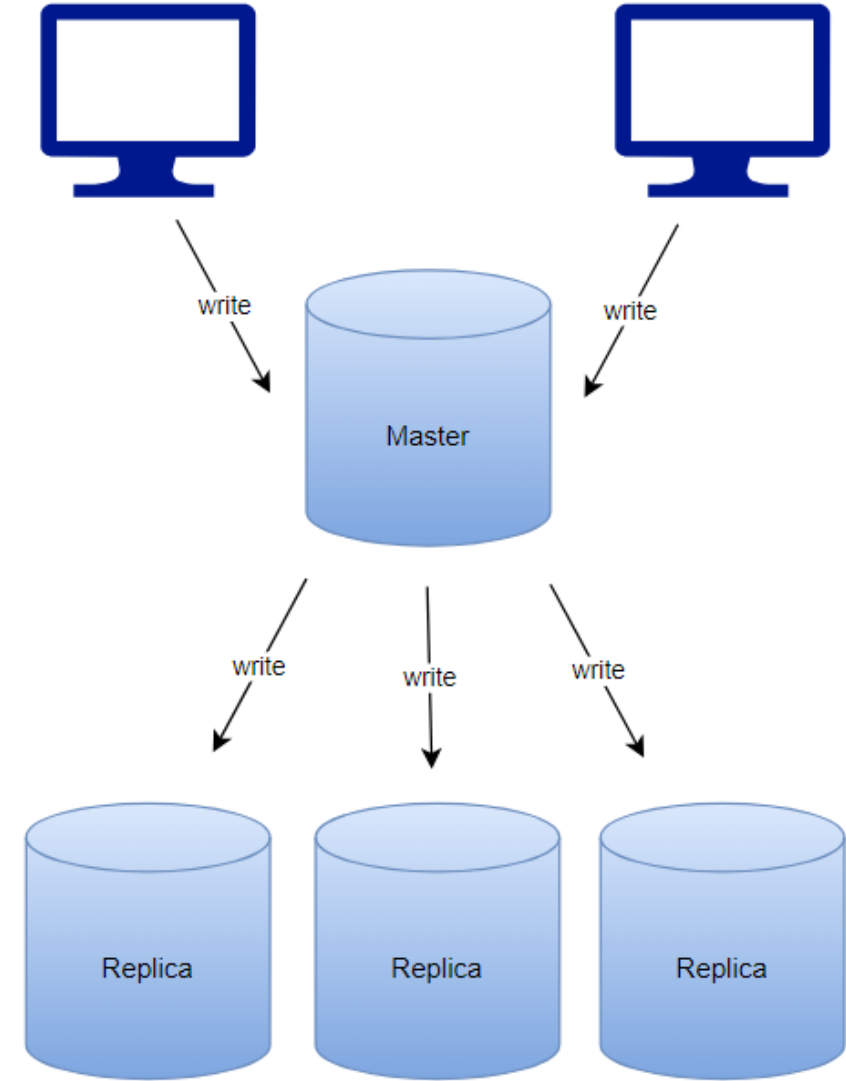
REPLICATION

- Replication is the process of copying and maintaining data from one location to another in real time or on a schedule.

Ensure availability and business continuity

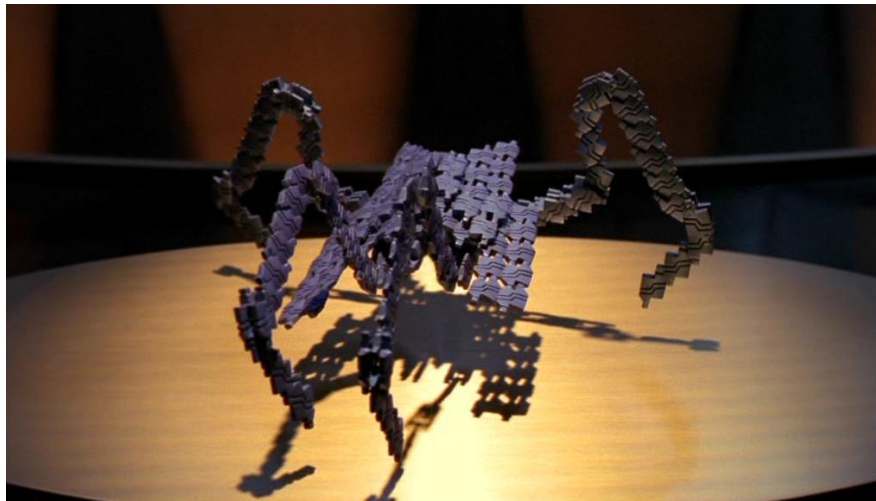
Provide disaster recovery options

Improve performance and accessibility

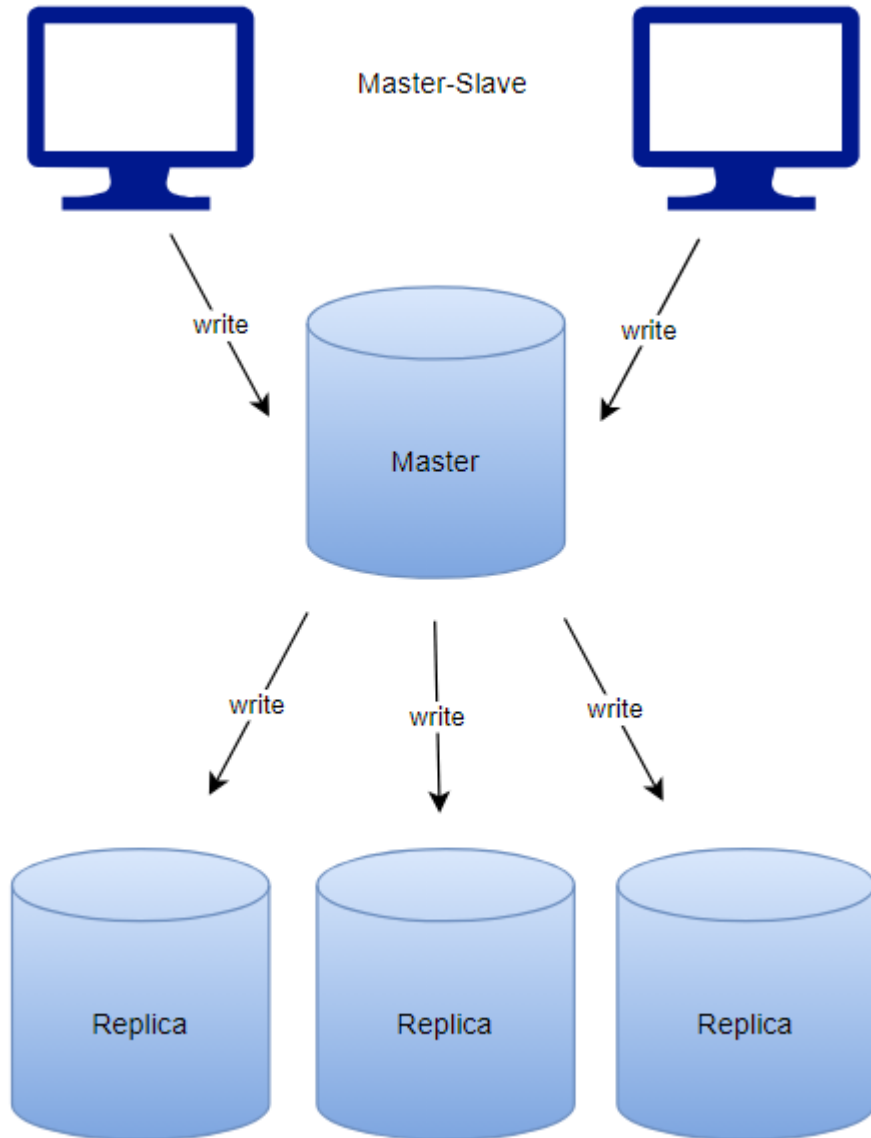


REPLICATION TYPES

Type	Description	Example
Synchronous	Writes data to primary & replica simultaneously	Zero data loss but higher latency
Asynchronous	Replica updated after write	Small lag, better performance
Geo-Replication	Data copied to remote site	Disaster recovery for regional outages

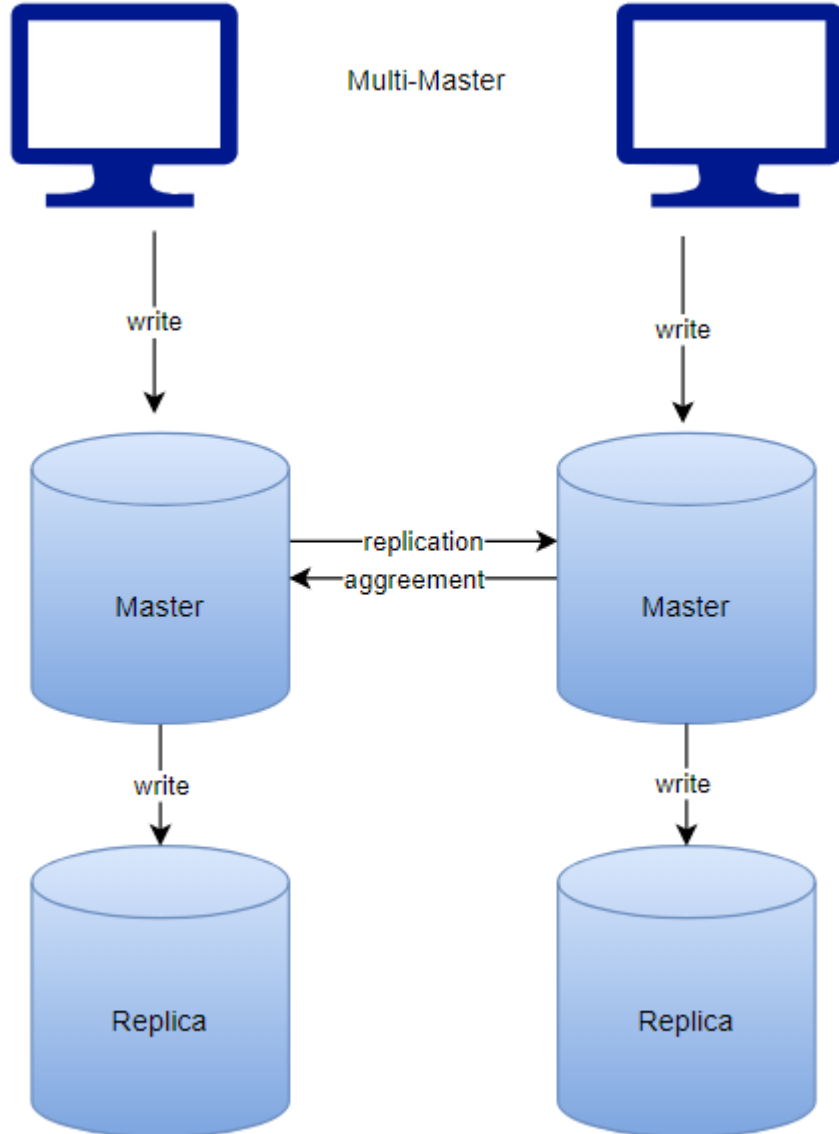


REPLICATION TOPOLOGIES



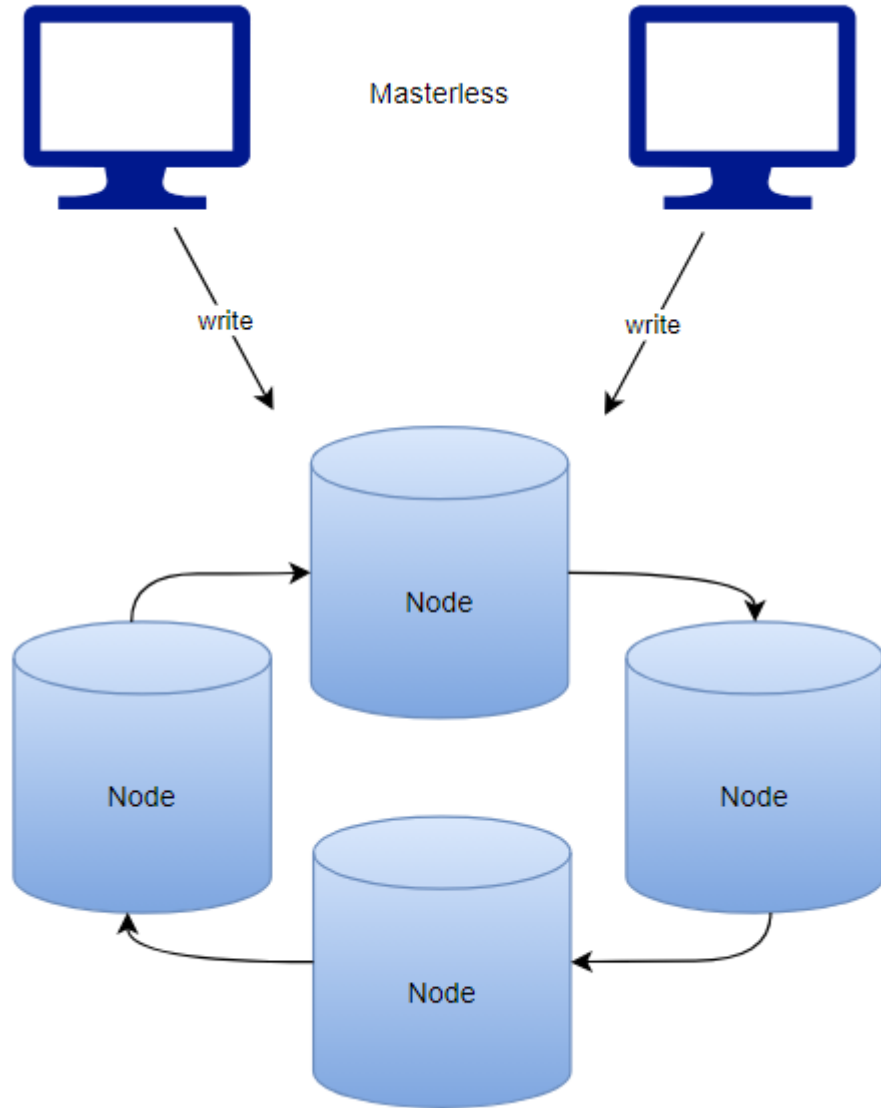
- One primary node (master) handles all writes.
- One or more secondary nodes (slaves) replicate data from the master.
- Slaves are usually read-only.
- If the master fails → manual or automatic failover to a slave.
- Ensures data consistency and clear hierarchy.

REPLICATION TOPOLOGIES



- Multiple nodes can handle both reads and writes.
- Changes are replicated across all nodes.
- Provides high availability and low latency for distributed users.
- Requires conflict detection and resolution.
- Ideal for global or active-active environments

REPLICATION TOPOLOGIES



- No single master.
- All nodes are equal peers.
- Each node can accept reads and writes.
- Consistency achieved through quorum agreement (e.g., majority vote).
- Designed for high availability and fault tolerance.

DISASTER RECOVERY

- Disaster Recovery (DR) is the process of restoring IT systems and data after a major disruption.
- Ensures business continuity when hardware, software, or facilities fail.
- Focuses on minimizing downtime and data loss.
- Includes plans, tools, people, and processes for recovery.
- Often part of a broader Business Continuity Plan (BCP).

DISASTER RECOVERY PLANNING

- DR Planning Components:
 - Business Impact Analysis (BIA): Criticality assessment
 - Risk Assessment: Threat identification and mitigation
 - Recovery Strategies: Technology and process plans
 - Testing and Validation: Regular DR exercises
 - Application Disaster Recovery Plan

DR SITE TYPE

- DR Site Types:
 - Hot Site: Fully equipped and staffed
 - Warm Site: Equipment ready, limited staffing
 - Cold Site: Space and power, no equipment
 - Cloud DR: Infrastructure as a Service

KEY DISASTER RECOVERY (DR) METRICS

Metric	Definition	Purpose
RTO (Recovery Time Objective)	Maximum time a system can be down after a disaster	Defines how quickly recovery must happen
RPO (Recovery Point Objective)	Maximum acceptable data loss in time	Defines how much data can be lost
RTA (Recovery Time Actual)	Actual time taken to recover during an incident or test	Measures DR performance against RTO
MTTR (Mean Time to Recover)	Average time to repair and restore over multiple incidents	Evaluates operational efficiency
Availability (%)	Percentage of uptime over a given period	Measures service reliability

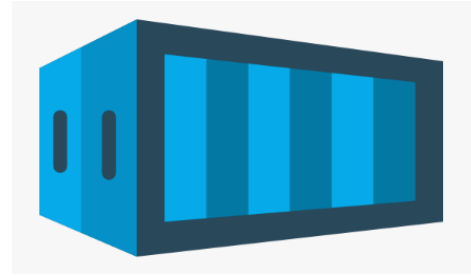
RECOVERY STRATEGIES



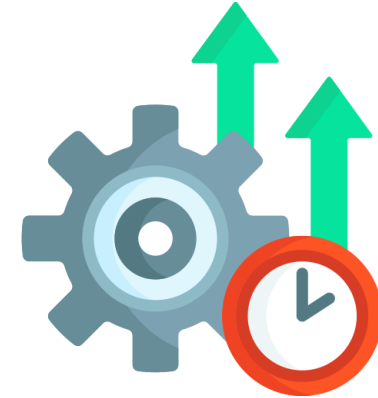
Backup Restore



VM Recovery



Container Recovery



Application Failover

CLOUD DATA PROTECTION

- Cloud Backup Strategies:
 - Direct-to-Cloud: Backup directly to cloud storage
 - Hybrid Approach: Local backup with cloud copy
 - Cloud-Native: Cloud-first backup architecture
- Pure Cloud Block Store:
 - Native AWS, Azure, Google Cloud integration
 - Consistent management across hybrid environments
 - Cloud disaster recovery capabilities
- Benefits and Considerations:
 - Unlimited scale and geographic distribution
 - Cost optimization with storage tiers
 - Network bandwidth and egress cost management

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LAB 5 : AWS SNAPSHOTS & AMIS

- Use AWS EBS for Volume Snapshots
- Use Amazon Machine Images for instance backups
- Restore from a Snapshot
- Launch new AMI instance